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IBM SKILLSBUILD / MACHINE LEARNING & APPLIED AI

SkillMatch AI

A practical skill-gap review for the role you want next.

SDG 4 · QUALITY EDUCATION

SDG 8 · DECENT WORK

PROJECT SUMMARY

SkillMatch AI is a lightweight Streamlit application that helps learners compare the skills they describe with a selected target role. It recognises terms from a small bundled catalogue, lists matched and not-yet-recognised role requirements, estimates text similarity with TF-IDF and cosine similarity, and presents a suggested learning roadmap. The deployed prototype can be tried online; the outputs are intended for reflection and planning, not recruitment decisions.

WHO IT SERVES

University students and early-career applicants

WHAT IT USES

Python, Streamlit, pandas, scikit-learn and bundled JSON catalogues

WHAT IT AVOIDS

Paid API dependency, large runtime dataset, and hiring-probability claims

LIVE PROTOTYPE

skillmatch-ai-rs.streamlit.app

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Status: deployed educational prototype; user validation and impact evaluation remain future work.

The problem and project objectives

The problem framing below is a hypothesis to validate with learners, not a claim based on a completed user study.

PROBLEM HYPOTHESIS

Students preparing for internships may see lengthy lists of role requirements without a clear way to compare them with their current experience. General course lists can leave the next learning step ambiguous. A transparent, role-focused checklist could help learners decide what to practise first.

What the prototype sets out to do

01

Make role requirements easier to scan

Recognise terms from a curated skill catalogue and separate matched from unrecognised role skills.

02

Connect comparison to action

Generate a role-specific roadmap with practical learning steps.

03

Keep the method understandable

Show how coverage and text similarity are calculated and explain where they can mislead.

User flow

01 Enter

Skills, tools, projects and experience

→

02 Choose

One of five role templates

→

03 Compare

Catalogue matches + text similarity

→

04 Plan

Review gaps and export next steps

CURRENT MVP SCOPE

- Text input for skills and experience; optional job-description input.
- Five technology-role templates: Data Analyst, Machine Learning Intern, Data Scientist, Frontend Developer and Backend Developer.
- Recognised skills, matched/missing requirements, coverage metric, similarity metric, learning roadmap, CSV and TXT downloads.
- Not included: a validated hiring predictor, live vacancy search, recruiter ranking, user accounts, or a verified credential system.
- Role requirements and roadmaps are curated and limited; users should verify against the original job listing.

How SkillMatch AI works

A compact rule-and-text pipeline; it does not train a predictive model or call a paid generative-AI service.

01 INPUT

Profile text, selected role and optional job description.

02 CATALOGUE

Bundled role and skill dictionaries (JSON).

03 ANALYSIS

Phrase matching plus TF-IDF and cosine similarity.

04 OUTPUT

Matched/missing skills, coverage, roadmap, CSV and TXT.

Implementation map

Layer	Current implementation	Purpose
Interface	Streamlit	Profile entry, role selection, display of the analysis and downloads.
Data handling	Python + pandas	Organises the report and exports analysis to CSV.
Skill recognition	Curated JSON catalogue + phrase matching	Finds catalogue-listed skill terms in text; it may miss synonyms or context.
Text comparison	scikit-learn TF-IDF + cosine similarity	Estimates overlap between the profile and role/job text; not a competence score.
Role definition	Bundled JSON role templates	Provides the required-skill list and roadmap for each supported role.

METRICS SHOWN IN THE APP

Skill coverage = recognised required skills divided by all required skills in the selected role, expressed as a percentage. Text similarity uses TF-IDF vectors and cosine similarity, shown on a 0-100 display scale. Neither metric estimates hiring probability. The optional job description is used for text comparison; it does not rewrite the curated list of missing role skills.

Intended contribution and validation plan

The project is designed to support learning choices; outcomes should be measured before impact claims are made.

04 **Quality education**

Make role requirements easier to interpret and turn them into practical self-directed learning steps.

08 **Decent work and economic growth**

Support career preparation by helping learners organise skill development for internships and entry-level roles.

How the prototype should be evaluated

Question	Method	Possible measure
Does the catalogue recognise skills correctly?	Manually label sample profiles and compare detected terms.	Precision, missed-term rate, synonym misses
Do learners understand the result?	Ask participants to complete a short role-gap task.	Task completion, time, comprehension check
Is the roadmap useful?	Short post-task rating and optional comments.	Clarity/usefulness rating, repeated themes
Are results consistent across roles?	Test multiple profiles against all five role templates.	Coverage edge cases, empty/ambiguous inputs

LIMITATIONS & SAFEGUARDS

- Small catalogue; synonyms, context and emerging tools can be missed.
- Mentioning a skill does not prove proficiency or project depth.
- Text similarity is lexical overlap, not hiring probability.
- Avoid sensitive personal information; cross-check the original job listing.

NEXT STEPS

1. **Review** — Test the deployed app with varied profiles and edge cases.
2. **Improve** — Add reviewed aliases and cite role-skill sources.
3. **Validate** — Run small user sessions; document feedback and error rates.
4. **Extend carefully** — Prioritise accessibility and clear privacy handling before new features.

LIVE PROTOTYPE · https://skillmatch-ai-rs.streamlit.app